

Vision + Language + Audio

Myntra's 2025 visual search lets users upload a saree photo AND ask "show me similar, under ₹3,000". Three modalities fused into one query. This is where UX is heading — and every frontier LLM in 2026 ships multimodal by default.

Which of your apps would be better if users could just show you what they mean?

L1

WHAT'S INSIDE

- 01** What is multimodal AI
- 02** CLIP — the seed of modern multimodal
- 03** Three flagship multimodal LLMs
- 04** Image + text search — 30 lines
- 05** Evaluation matters more here

WHY THIS LESSON MATTERS

Users don't think in modalities. They upload AND ask.

WHY

Real

A real reason this matters in industry today.

SCALE

Today

A real reason this matters in industry today.

SPEED

Now

A real reason this matters in industry today.

IMPACT

Yours

A real reason this matters in industry today.

LEARNING OUTCOMES

By the end of this lesson, you will be able to —

1

Define multimodal AI + 3 production systems.

2

Explain contrastive training (CLIP).

3

Use GPT-4o/Claude/Gemini vision APIs.

4

Image+text search over 10k items.

5

Pick metrics for multimodal output.

AGENDA

What we'll cover today

Short, sequenced parts. Each one builds on the last.

01

What is multimodal AI

Models that jointly process 2+ input channels

02

CLIP — the seed of modern...

One embedding space for image + text

03

Three flagship multimodal...

Key concept of this lesson.

04

Image + text search — 30...

open_clip + ChromaDB

01

PART 1 OF 4

What is multimodal AI

Models that jointly process 2+ input channels

What is multimodal AI

Text + image → describe, answer, extract.

Text + audio → transcribe, translate, summarise.

Text + video → scene description, chapter segmentation.

Audio + video → lip-sync, person identification.

DEFAULT

Multimodal

Text-only is a historical footnote by 2026.

DEFINITION

What is multimodal AI

Models that jointly process 2+ input channels

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 2 OF 4

CLIP — the seed of modern multimodal

400M (image, caption) pairs from the web.

Contrastive loss: matched close, mismatched far.

Zero-shot classification: "photo of a CAT" vs "photo of a DOG".

Foundation for DALL-E, Stable Diffusion, GPT-4o vision.

IDEA

Shared space

Image and text live on the same grid.

KEY POINTS

- 400M (image, caption) pairs from the web
- Contrastive loss: matched close, mismatched far
- Zero-shot classification: "photo of a CAT" vs "photo of a DOG"
- Foundation for DALL-E, Stable Diffusion, GPT-4o vision

CONCEPT 3 OF 4

Three flagship multimodal LLMs

A core idea you'll use throughout this lesson.

DEFINITION

Three flagship multimodal LLMs

EXAMPLE

Real-world example: applies directly to Indian industry contexts.

CONCEPT 4 OF 4

Image + text search — 30 lines

Encode every image with CLIP image encoder.

Query: encode text; cosine to image embeddings.

Top-k most similar → the search result.

Cache embeddings in S3 parquet. Re-embed only on schema change.

BUILD

30 lines

Image search in one lecture.

KEY POINTS

- Encode every image with CLIP image encoder
- Query: encode text; cosine to image embeddings
- Top-k most similar → the search result
- Cache embeddings in S3 parquet. Re-embed only on schema change

02

PART 2 OF 4

Engage and reflect

Pause, talk, and connect what you just learned to your own work.

✦ THINK • PAIR •
SHARE

ACTIVITY

Pick the right multimodal stack

PROMPT

Pick the right multimodal stack

HOW TO RUN IT

- 1 Pick the right multimodal stack
- 2
- 3
- 4 Reveal: instructor confirms the canonical answer.

REFLECT & APPLY

Three questions before you close this lesson

Spend 60 seconds on each. Write the answers down — no scrolling, no shortcuts.

Q1

Which modality do YOUR users generate most?

Q2

Where would image search beat keyword search in your work?

Q3

What's the biggest UX lift a multimodal upgrade would give?

03

PART 3 OF 4

Knowledge check

Three short questions. One minute each. Honest answers only.

CLIP training objective:

A

Supervised

B

Contrastive

C

Autoregressive

D

RL

CLIP training objective:



CORRECT ANSWER

Contrastive

WHY

Pull matched image-text pairs closer; push mismatched apart

Best 2026 model for native video understanding:

A

GPT-4o

B

Claude 3.5

C

Gemini 2

D

DALL-E

Best 2026 model for native video understanding:



CORRECT ANSWER

Gemini 2

WHY

Gemini 2 is natively video-aware

Zero-shot with CLIP means:

A

No training

B

Embed candidate text classes; pick closest to image

C

Use labels

D

One-shot

Zero-shot with CLIP means:



CORRECT ANSWER

Embed candidate text classes; pick closest to image

WHY

Class names encoded as text; closest to the image embedding wins

04

PART 4 OF 4

Apply and close

One thing to remember. One thing to ship this week.

SUMMARY

Five sentences to remember

If you remember nothing else from this lesson, remember these five lines.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

WORKED EXAMPLE

How this lesson plays out in one real Indian context

THE SCENARIO

Text + image → describe, answer, extract.

Text + audio → transcribe, translate, summarise.

Text + video → scene description, chapter segmentation.

Audio + video → lip-sync, person identification.

DEFAULT

Multimodal

Text-only is a historical footnote by 2026.

THE TAKEAWAY

"The users of 2026 expect all four."

WHAT TO NOTICE

1

What is multimodal AI

2

CLIP — the seed of modern...

3

Three flagship multimodal...

4

Image + text search — 30...

Image search over your photos

100 personal photos; natural-language queries.

DELIVERABLES

- Encode with open_clip.
- Store in ChromaDB.
- Query: "me at the beach at sunset".
- Report hit@5.

WHAT 'GOOD' LOOKS LIKE

Deliverable: Repo + hit@5 number.

ONE THING TO REMEMBER

“

"The users of 2026 expect all four."

— AI Learning Hub

END OF LESSON 7

What you can now do

Each one of these is now part of your working vocabulary.

- 1 Pick the right tool — never the loudest one.
- 2 Practice the framework on a real example.
- 3 Watch for the three classic pitfalls.
- 4 Document your decision in one sentence.
- 5 Carry the discipline forward to the next lesson.

NEXT → Lesson 2 — Speech AI.